NCERT Summary: Topic 08: Atmospheric Circulation & Weather Systems — Raw Notes

Source Integration

- NCERT Class 7 Our Environment, Chapter 4: Air (pressure belts and wind portions)[cite: 8]
- NCERT Class 11 Fundamentals of Physical Geography, Chapter 9: Atmospheric Circulation and Weather Systems[cite: 8]

Core Factual & Conceptual Architecture

1. Atmospheric Pressure & Barometric Distribution

- Definition: Atmospheric pressure represents the weight of an air column resting on a unit surface area, measured from mean sea level to the upper boundary of the atmosphere[cite: 8].
- Measurement Units: Expressed in millibars (mb), with average sea-level pressure standardized at 1,013.2 mb using mercury or aneroid barometers[cite: 8].
- Vertical Gradient: Air pressure is highest at sea level and decreases rapidly with altitude (averaging roughly 1 mb per 10 meters near the surface, dropping to ~540 mb at 5 km and ~265 mb at 10 km)[cite: 8].
- Isobars: Cartographic lines drawn on weather charts connecting locations with identical mean sea-level pressure[cite: 8].
- Pressure-Weather Correlations:
  - Low-Pressure Systems: Characterized by warm, ascending air parcels, low atmospheric density, cloudy skies, and wet precipitation[cite: 8].
  - High-Pressure Systems: Characterized by cold, dense, subsiding air parcels, clear skies, and calm weather[cite: 8].

2. Forces Governing Wind Velocity and Direction

- Pressure Gradient Force (PGF): The force generated by horizontal pressure differences across distance[cite: 8]. The PGF is strong when isobars are closely spaced and weak when isobars are widely separated[cite: 8].
- Frictional Force: Frictional drag exerted by surface irregularities (mountains, trees, buildings) that slows down wind speed[cite: 8]. Effective primarily in the lower 1 to 3 kilometers of the troposphere and negligible over open oceans[cite: 8].
- Coriolis Force: An apparent deflection force resulting from the Earth's axial rotation[cite: 8]. Deflects moving air and water to the **right in the Northern Hemisphere** and to the **left in the Southern Hemisphere**[cite: 8]. It is **zero at the equator** and **maximum at the poles**, and is directly proportional to both latitude and wind velocity[cite: 8].
- Gravity: Acts vertically downward, balancing the much larger vertical pressure gradient force and preventing rapid vertical air displacement[cite: 8].
- Geostrophic Wind: When isobars are straight and surface friction is absent, the pressure gradient force balances the Coriolis force, causing the wind to blow **parallel to the isobars** (typically observed in the upper troposphere at 2 to 3 km elevation)[cite: 8].
- Cyclonic vs. Anticyclonic Circulation:
  - Cyclones: Low-pressure centers featuring counterclockwise wind convergence in the Northern Hemisphere and clockwise convergence in the Southern Hemisphere[cite: 8].
  - Anticyclones: High-pressure centers featuring clockwise wind divergence in the Northern Hemisphere and counterclockwise divergence in the Southern Hemisphere[cite: 8].

3. World Pressure Belts & The General Circulation

- Global Pressure Belts:
  - Equatorial Low (Doldrums): Thermally induced low-pressure belt near the equator driven by intense insolation and rising air[cite: 8].
  - Sub-Tropical Highs (Horse Latitudes): Dynamically induced high-pressure belts located around 30°N and 30°S due to the subsidence of upper-air currents[cite: 8].
  - Sub-Polar Lows: Dynamically induced low-pressure belts located around 60°N and 60°S where warm westerlies meet cold polar easterlies[cite: 8].
  - Polar Highs: Thermally induced high-pressure caps situated over both poles due to extreme surface cooling[cite: 8].
- The Three-Cell Atmospheric Circulation Model:
  - Hadley Cell: Driven by thermal heating at the Inter-Tropical Convergence Zone (ITCZ); warm air rises, flows poleward aloft, and subsides at the subtropical highs, returning near the surface as the northeast/southeast trade winds[cite: 8].
  - Ferrel Cell: Mid-latitude thermal circulation cell driven by surface westerlies and upper-air return

currents[cite: 8].
  - Polar Cell: High-latitude thermal loop where cold polar air sinks and flows equatorward as polar easterlies[cite: 8].
- Planetary Surface Wind Belts:
  - Trade Winds (0° to 30°): Persistent easterly surface winds blowing toward the equator[cite: 8].
  - Prevailing Westerlies (30° to 60°): Persistent westerly surface winds blowing toward the poles[cite: 8].
  - Polar Easterlies (60° to 90°): Cold easterly surface winds blowing from polar highs toward sub-polar lows[cite: 8].

4. Local Winds & Thermal Circulations

- Sea & Land Breezes: Diurnal thermal winds; sea breezes blow from ocean to land during warm daytime hours, while land breezes reverse the flow from land to ocean at night[cite: 8].
- Valley & Mountain Breezes: Mountain slopes heat rapidly by day, drawing valley air upslope (valley breeze); at night, dense cooled air drains downslope into valleys (mountain breeze or katabatic wind) [cite: 8].
- Foehn and Chinook Winds: Warm, dry leeward winds descending mountain barriers after losing moisture on windward slopes, capable of rapidly melting snow cover[cite: 8].

5. Air Masses and Fronts

- Air Mass: A massive body of homogeneous air spanning hundreds of kilometers whose physical properties (temperature and humidity) remain uniform horizontally, formed over stable source regions[cite: 8].
- Fronts: The active boundary zone separating two distinct air masses with contrasting temperatures and densities (frontogenesis)[cite: 8].
  - Cold Front: Occurs when advancing cold air actively wedges beneath and displaces warm air[cite: 8].
  - Warm Front: Occurs when warm air glides up and over a retreating wedge of cold air[cite: 8].
  - Stationary Front: Occurs when the boundary between two air masses stalls entirely[cite: 8].
  - Occluded Front: Occurs when a fast-moving cold front overtakes a warm front, completely lifting the warm air mass off the ground[cite: 8].

6. Extra-Tropical and Tropical Cyclones

- Extra-Tropical (Mid-Latitude) Cyclones:
  - Form along the polar front in mid-to-high latitudes through wave development along a stationary front[cite: 8].
  - Characterized by distinct warm and cold fronts, larger areal coverage, lower wind velocities, and west-to-east movement[cite: 8].
- Tropical Cyclones:
  - Violent, destructive low-pressure storms originating exclusively over warm tropical oceans where sea surface temperatures exceed 27°C[cite: 8].
  - Global Nomenclature: Cyclones in the Indian Ocean, Hurricanes in the Atlantic, Typhoons in the Western Pacific, and Willy-willies in Australia[cite: 8].
  - Structural Anatomy: A calm central "eye" of subsiding air (150–250 km wide) surrounded by an intense "eye wall" featuring towering cumulonimbus clouds, torrential rainfall, and maximum destructive wind velocities (exceeding 250 km/h)[cite: 8].
  - Energetics & Landfall: Powered continuously by latent heat of condensation released from warm tropical seas[cite: 8]. Upon crossing a coastline (landfall), the moisture supply is cut off, causing the storm to rapidly dissipate[cite: 8]. Real-world case study: The 1999 Odisha Super Cyclone[cite: 8].

7. Localized Convective Storms: Thunderstorms and Tornadoes

- Thunderstorms: Short-duration, violent convective storms driven by intense daytime heating and moisture, marked by cumulonimbus clouds, lightning, thunder, and potential hail[cite: 8].
- Tornadoes: Violently rotating, funnel-shaped columns of descending air associated with severe thunderstorms, characterized by extremely low central pressure and localized catastrophic destruction[cite: 8].

✍ Notes & Updates